

PLSC 504 – Fall 2026

Regression Models for Binary and Nominal Responses

August 31, 2026

Binary Outcomes: Quick Review

Latent:

$$Y_i^* = \mathbf{X}_i\beta + u_i$$

Observed:

$$Y_i = 0 \text{ if } Y_i^* < 0$$

$$Y_i = 1 \text{ if } Y_i^* \geq 0$$

So:

$$\begin{aligned}\Pr(Y_i = 1) &= \Pr(Y_i^* \geq 0) \\ &= \Pr(\mathbf{X}_i\beta + u_i \geq 0) \\ &= \Pr(u_i \geq -\mathbf{X}_i\beta) \\ &= \Pr(u_i \leq \mathbf{X}_i\beta) \\ &= \int_{-\infty}^{\mathbf{X}_i\beta} f(u) du\end{aligned}$$

“Standard logistic” PDF:

$$\Pr(u) \equiv \lambda(u) = \frac{\exp(u)}{[1 + \exp(u)]^2}$$

CDF:

$$\begin{aligned}\Lambda(u) &= \int \lambda(u) du \\ &= \frac{\exp(u)}{1 + \exp(u)} \\ &= \frac{1}{1 + \exp(-u)}\end{aligned}$$

Logistic $\rightarrow$ "Logit"

$$\begin{aligned}\Pr(Y_i = 1) &= \Pr(Y_i^* > 0) \\ &= \Pr(u_i \leq \mathbf{X}_i\beta) \\ &= \Lambda(\mathbf{X}_i\beta) \\ &= \frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)}\end{aligned}$$

$$\text{(equivalently)} = \frac{1}{1 + \exp(-\mathbf{X}_i\beta)}$$

$$L = \prod_{i=1}^N \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right)^{Y_i} \left[1 - \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right) \right]^{1-Y_i}$$

$$\ln L = \sum_{i=1}^N Y_i \ln \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right) + (1 - Y_i) \ln \left[1 - \left(\frac{\exp(\mathbf{X}_i\beta)}{1 + \exp(\mathbf{X}_i\beta)} \right) \right]$$

Normal $\rightarrow$ “Probit”

$$\begin{aligned}\Pr(Y_i = 1) &= \Phi(\mathbf{X}_i\boldsymbol{\beta}) \\ &= \int_{-\infty}^{\mathbf{X}_i\boldsymbol{\beta}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(\mathbf{X}_i\boldsymbol{\beta})^2}{2}\right) d\mathbf{X}_i\boldsymbol{\beta}\end{aligned}$$

$$L = \prod_{i=1}^N [\Phi(\mathbf{X}_i\boldsymbol{\beta})]^{Y_i} [1 - \Phi(\mathbf{X}_i\boldsymbol{\beta})]^{(1-Y_i)}$$

$$\ln L = \sum_{i=1}^N Y_i \ln \Phi(\mathbf{X}_i\boldsymbol{\beta}) + (1 - Y_i) \ln [1 - \Phi(\mathbf{X}_i\boldsymbol{\beta})]$$

Logit and Probit, Explained

Things we talked about in PLSC 503:

- Odds ratios and the random utility model
- Model estimation and interpretation
- Marginal effects, predictions, etc.
- Assessing model fit
- A couple variants (e.g., c-log-log)

Extensions: Two Topics, One Theme

Things:

- Models for dealing with “separation”
- Models for *rare events*

Common Focus: Shortage of information on Y

“Separation” = “perfect prediction” = “monotone likelihood”

Intuition: House votes on the PPACA (3/21/2010):

	Dems	
Yeas	0	1
0	178	34
1	0	219

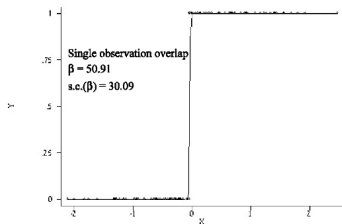
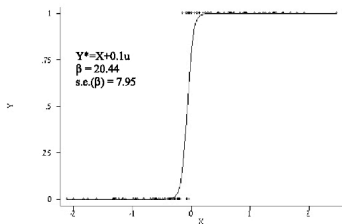
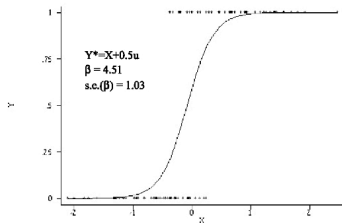
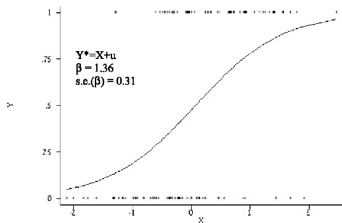
$\Pr(Y = 1|X = 0) = ?$

“Separation” means that:

- $\hat{\beta}_X = \pm\infty$
- $\widehat{\text{s.e.}}_{\beta} = \infty$
- $\left. \frac{\partial^2 \ln L}{\partial X^2} \right|_{\hat{\beta}} = 0$ (monotone likelihood)

Separation Illustrated

Figure 1: Actual and Predicted Values, Simulated Logistic Regressions



Separation: What Happens

```
> set.seed(7222009)
> Z<-rnorm(500)
> W<-rnorm(500)
> Y<-rbinom(500,size=1,prob=plogis((0.2+0.5*W-0.5*Z)))
> X<-rbinom(500,1,(pnorm(Z)))
> X<-ifelse(Y==0,0,X) # Induce separation of Y on X

> summary(glm(Y~W+Z+X,family="binomial"))

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.638      0.133   -4.81  1.5e-06 ***
W              0.653      0.140    4.67  3.0e-06 ***
Z             -1.134      0.146   -7.76  8.3e-15 ***
X             20.915     861.458    0.02   0.98
---
Number of Fisher Scoring iterations: 18

# Change the maximum # of iterations / convergence tolerance:

> summary(glm(Y~W+Z+X,family="binomial",maxit=100,epsilon=1e-16))

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.638      0.133   -4.81  1.5e-06 ***
W              0.653      0.140    4.67  3.0e-06 ***
Z             -1.134      0.146   -7.76  8.3e-15 ***
X             34.915  5978532.779    0.00      1
---
Number of Fisher Scoring iterations: 32

Warning message:
glm.fit: fitted probabilities numerically 0 or 1 occurred
```

One Solution: Exact Logistic Regression

Exact logistic regression (ELR):

- Cox (1970, Ch. 4); Hirji et al. (1987 *JASA*); Mehta & Patel (1995 *Stat. Med.*); Forster et al. (2003 *Stat. & Comp.*); Zamar and Graham (2007 *J. Stat. Soft.*).
- Conditions on permutations of covariate patterns
- $\longrightarrow$ Always has finite solutions for $\hat{\beta}$
- Implementation:
 - `elrm` in R ; `exlogistic` in Stata
 - Fitted via MCMC; see Forster et al. for details
 - In practice, there are often computational issues...

Firth's (1993) Correction

Firth proposed:

$$L(\beta|Y)^* = L(\beta|Y) |\mathbf{I}(\beta)|^{\frac{1}{2}}$$

$$\ln L(\beta|Y)^* = \ln L(\beta|Y) + 0.5 \ln |\mathbf{I}(\beta)|$$

“Penalized likelihood”:

- Is consistent
- Eliminates small-sample bias
- Exist given separation
- To Bayesians, it's “Jeffreys' prior”:

$$P(\theta) = \sqrt{\det [I(\theta)]}$$

- “Profile” (= “concentrated”) likelihood
- $\hat{\beta}$ can be asymmetrical...
- $\rightarrow$ can affect “normal” inference...
- Plotting the profile likelihood and calculating alternative C.I.s is recommended

Two directions:

- R
 - `elrm` (exact logistic regression via MCMC)
 - `brlr` (“bias-reduced logistic regression”)
 - `logistf` (“Firth’s logistic regression”)
- Stata
 - `exlogistic` (exact logistic regression)
 - `firthlogit` (Firth corrected logit)

Example: Pets as Family

Some data, and a silly question:

- CBS/NYT Poll, April 1997
- Standard political/demographics, plus
- “Do you consider your pet to be a member of your family, or not?”
- Yes = 84.4%, No = 15.6%

Data:

```
> summary(Pets)
```

petfamily	female	married	partyid	education
Min. :0.000	Min. :0.000	Married :442	Democrat :225	< HS : 71
1st Qu.:1.000	1st Qu.:0.000	Widowed : 46	Independent:214	HS diploma :244
Median :1.000	Median :1.000	Divorced/Sep:118	GOP :229	Some college:184
Mean :0.844	Mean :0.556	NBM :118	NA's : 58	College Grad:131
3rd Qu.:1.000	3rd Qu.:1.000	NA's : 2		Post-Grad : 96
Max. :1.000	Max. :1.000			

Pets as Family: Basic Model

```
> Pets.1<-glm(petfamily~female+as.factor(married)+as.factor(partyid)
+             +as.factor(education),data=Pets,family=binomial)
> summary(Pets.1)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.0133	0.5388	3.74	0.00019	***
femaleMale	-0.6959	0.2142	-3.25	0.00116	**
as.factor(married)Married	-0.0657	0.2911	-0.23	0.82147	
as.factor(married)NBM	0.4599	0.3957	1.16	0.24504	
as.factor(married)Widowed	-0.1568	0.4921	-0.32	0.75007	
as.factor(partyid)Democrat	-0.1241	0.4286	-0.29	0.77213	
as.factor(partyid)GOP	-0.0350	0.4321	-0.08	0.93537	
as.factor(partyid)Independent	-0.1521	0.4299	-0.35	0.72338	
as.factor(education)College Grad	0.2511	0.4121	0.61	0.54228	
as.factor(education)HS diploma	0.0595	0.3685	0.16	0.87182	
as.factor(education)Post-Grad	0.1946	0.4331	0.45	0.65321	
as.factor(education)Some college	0.0587	0.3867	0.15	0.87928	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Null deviance: 627.14 on 723 degrees of freedom
Residual deviance: 612.76 on 712 degrees of freedom
AIC: 636.8

Number of Fisher Scoring iterations: 4



Pets as Family: More Complicated Model

```
> Pets.2<-glm(petfamily~female+as.factor(married)*female+as.factor(partyid)+  
+ as.factor(education),data=Pets,family=binomial)
```

```
> summary(Pets.2)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.2971	0.6166	3.73	0.0002	***
femaleMale	-1.1833	0.5305	-2.23	0.0257	*
as.factor(married)Married	-0.3218	0.4470	-0.72	0.4716	
as.factor(married)NBM	0.1854	0.6140	0.30	0.7628	
as.factor(married)Widowed	-0.7415	0.5780	-1.28	0.1995	
as.factor(partyid)Democrat	-0.1575	0.4297	-0.37	0.7140	
as.factor(partyid)GOP	-0.0445	0.4334	-0.10	0.9182	
as.factor(partyid)Independent	-0.1757	0.4312	-0.41	0.6837	
as.factor(education)College Grad	0.2332	0.4137	0.56	0.5730	
as.factor(education)HS diploma	0.0558	0.3703	0.15	0.8801	
as.factor(education)Post-Grad	0.2171	0.4342	0.50	0.6171	
as.factor(education)Some college	0.0358	0.3890	0.09	0.9266	
femaleMale:as.factor(married)Married	0.4853	0.5908	0.82	0.4114	
femaleMale:as.factor(married)NBM	0.5260	0.8051	0.65	0.5136	
femaleMale:as.factor(married)Widowed	15.2516	549.3719	0.03	0.9779	

Null deviance: 627.14 on 723 degrees of freedom
Residual deviance: 607.42 on 709 degrees of freedom
AIC: 637.4

Number of Fisher Scoring iterations: 14

What's Going On?

```
> with(Pets, xtabs(~petfamily+as.factor(married)+female))
```

```
, , female = Female
```

```
      as.factor(married)
petfamily Divorced/Sep Married NBM Widowed
      0           7      28   5      7
      1          67     199  58     32
```

```
, , female = Male
```

```
      as.factor(married)
petfamily Divorced/Sep Married NBM Widowed
      0           11      47   8      0
      1          33     168  47      7
```

Pets as Family: Firth Model

```
> Pets.Firth<-logistf(petfamily~female+
+                   as.factor(married)*female+as.factor(partyid)+
+                   as.factor(education),data=Pets,flic=TRUE)

> summary(Pets.Firth)
logistf(formula = petfamily ~ female + as.factor(married) * female +
        as.factor(partyid) + as.factor(education), data = Pets, flic = TRUE)
```

Model fitted by Penalized ML

Coefficients:

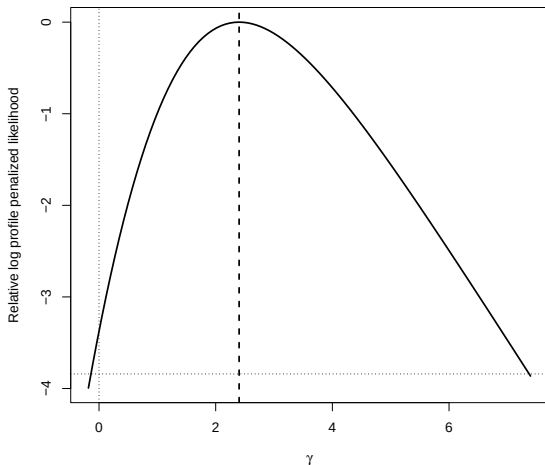
	coef	se(coef)	lower 0.95	upper 0.95	Chisq	p	method
(Intercept)	2.21205	0.608	1.019	3.405	16.17636	0.0000577	1
femaleMale	-1.13866	0.526	-2.187	-0.145	5.04186	0.0247420	2
as.factor(married)Married	-0.27387	0.442	-1.192	0.531	0.41518	0.5193531	2
as.factor(married)NBM	0.15888	0.601	-0.991	1.367	0.07322	0.7867048	2
as.factor(married)Widowed	-0.72627	0.572	-1.839	0.384	1.67233	0.1959467	2
as.factor(partyid)Democrat	-0.11818	0.425	-0.992	0.661	0.08159	0.7751592	2
as.factor(partyid)GOP	-0.00776	0.429	-0.888	0.780	0.00034	0.9852893	2
as.factor(partyid)Independent	-0.13643	0.427	-1.013	0.646	0.10813	0.7422784	2
as.factor(education)College Grad	0.23904	0.411	-0.574	1.024	0.34480	0.5570689	2
as.factor(education)HS diploma	0.07531	0.368	-0.667	0.763	0.04289	0.8359331	2
as.factor(education)Post-Grad	0.21837	0.433	-0.627	1.050	0.26307	0.6080189	2
as.factor(education)Some college	0.05240	0.387	-0.721	0.781	0.01888	0.8906980	2
femaleMale:as.factor(married)Married	0.45582	0.588	-0.661	1.613	0.63550	0.4253467	2
femaleMale:as.factor(married)NBM	0.52329	0.794	-1.023	2.050	0.45133	0.5017022	2
femaleMale:as.factor(married)Widowed	2.40167	1.722	-0.139	7.374	3.37453	0.0662116	2

Method: 1-Wald, 2-Profile penalized log-likelihood, 3-None

Likelihood ratio test=17.3 on 14 df, p=0.242, n=724

Wald test = 263 on 14 df, p = 0

Profile Likelihood Plot



Note: Plot shows estimated profile likelihood for different values of the parameter estimate for the interaction term `femaleMale:as.factor(married)Widowed`. Horizontal dotted line is the likelihood associated with $P \leq 0.05$.

Vertical dashed line is $\hat{\gamma}$; vertical dotted line indicates $\hat{\gamma} = 0$.

To summarize:

- Separation is an *estimation* problem...
- Separation $\rightarrow$ dropping covariates!
- Firth's approach $>$ ELR
- Can also be applied to other sparse-data situations:
 - “Fixed effects” logit models ([Cook et al. 2020](#))
 - Multinomial logit ([Cook et al. 2018](#))
 - Survival models ([Anderson et al. 2020](#))

Finally: Read [this twitter thread](#) before it's gone.

If events (“1s”) are rare, we can...

- Collect lots of “0s” for a few “1s”
- → Classification bias...

Example: Suppose that:

$$\Pr(Y_i) = \Lambda(0 + 1X_i)$$

then:

$$E(\hat{\beta}_0 - \beta_0) \approx \frac{\bar{\pi} - 0.5}{N\bar{\pi}(1 - \bar{\pi})}$$

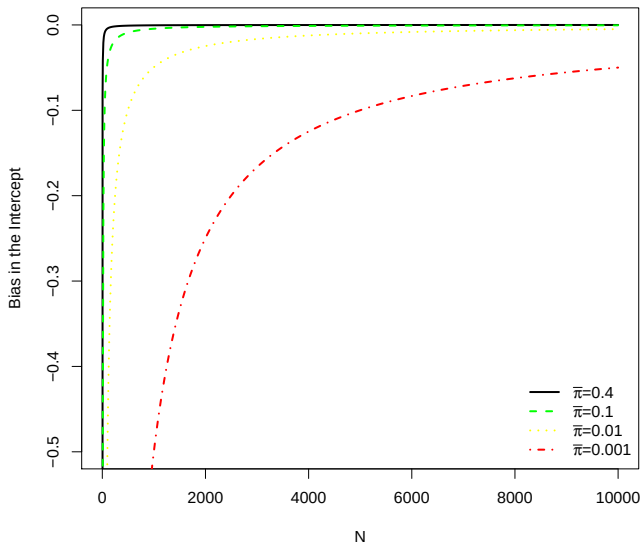
where $\bar{\pi} = \overline{\Pr(Y = 1)}$ is < 0.5 .

Bias is:

- always negative,
- worse as $\bar{\pi} \rightarrow 0$ (for fixed N),
- disappearing as $N \rightarrow \infty$.

Implication: *Logit/probit “work best” around $\bar{\pi} = 0.5$.*

Rare Event Bias, Illustrated



Alternatively, we can:

- Calculate $\tau = \frac{N_1 s}{N}$
- Collect data on all “1s”
- Sample from the “0s”
- Estimate a logit*
- *Correct* the estimates ex post...

*Yes, it *has* to be a logit, not a probit or something else...

Sampling and Weighting

Sampling...

- τ = fraction of “1s” in the population
- $\bar{Y}$ = fraction of ‘1s’ in the sample
- K&Z suggest $\bar{Y} \in [0.2, 0.5]$

Weighting...

$$w_1 = \frac{\tau}{\bar{Y}} \quad (\text{weights for “1s”})$$

$$w_0 = \frac{1 - \tau}{1 - \bar{Y}} \quad (\text{weights for “0s”})$$

$$\ln L(\beta | Y) = \sum_{i=1}^N w_1 Y_i \ln \Lambda(\mathbf{X}_i \beta) + w_0 (1 - Y_i) \ln [1 - \Lambda(\mathbf{X}_i \beta)]$$

Weighting:

- Good under (possible) misspecification, but
- Not as efficient as “prior correction,” and
- Gets s.e.s wrong...

Case-Control Data: Prior Correction

The prior correction is:

$$\hat{\beta}_{0pc} = \hat{\beta}_0 - \ln \left[\left(\frac{1 - \tau}{\tau} \right) \left(\frac{\bar{Y}}{1 - \bar{Y}} \right) \right]$$

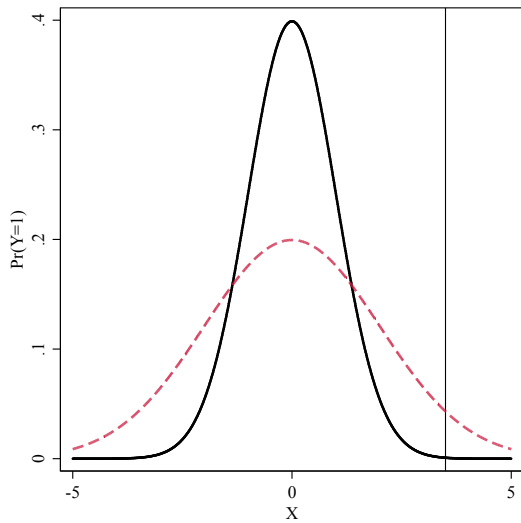
$$\text{bias}(\hat{\beta}) = (\mathbf{X}'\mathbf{W}\mathbf{X})^{-1}\mathbf{X}'\mathbf{W}\xi$$

where $\xi = f[w_i, \hat{\pi}_i, \mathbf{X}]$.

Correction is

$$\tilde{\beta} = \hat{\beta} - \text{bias}(\hat{\beta})$$

- Bias correction introduces additional variability...
- Ignoring it yields underpredictions (again).



Post-Correction Adjustments

Use:

$$\Pr(Y_i = 1) \approx \tilde{\pi}_i + C_i$$

where

$$C_i = (0.5 - \tilde{\pi}_i)\tilde{\pi}_i(1 - \tilde{\pi}_i)\mathbf{X}_i\mathbf{V}(\tilde{\beta})\mathbf{X}_i'$$

Puhr et al. (2017) note that Firth's method induces bias (toward 0.5) in predicted probabilities, and that the bias is worse when the baseline $\Pr(Y_i = 1)$ is low.

They introduce two modifications to deal with this:

- “Firth's logit with intercept correction” (FLIC)
- “Firth's logit with added covariate” (FLAC)

Through simulations, they show that both remove the bias; they have a slight preference for FLAC, but note that both work well relative to unmodified Firth regression. See also [here](#) for a more recent analysis of using Firth's methods for rare events.

Different data:

- Washington University's [American Panel Study](#) (TAPS)
- $N \approx 1000$ U.S. respondents, 2012-2017
- Outcome: "During the past year, have you ever run out of gas while driving a car or other vehicle?" (`RunOutOfGas`; 0=no, 1=yes)
- Predictors:
 - `Education` – twelve-category ordinal variable with values ranging from 3 to 15;
 - `Income` – a 15-category ordinal variable (each unit roughly corresponds to an increase of \$10,000 in annual income);
 - `Age` in years, as of 2016 (divided by 10);
 - `Female` – a binary indicator of sex, naturally-coded;
 - `Racial` classifications – binary variables for `White`, `Black`, and `Asian` identification;
 - Binary political party variables for `Democrat` and `GOP`; and
 - `Ideology` – a seven-point Likert variable, higher values indicate greater political conservatism

Basic Logit...

```
> table(TAPS$RunOutOfGas)

 0    1
943  28

> prop.table(table(TAPS$RunOutOfGas))

      0      1
0.9712 0.0288

> ROGLogit<-glm(RunOutOfGas~Education+Age10+Female+White+Black+Asian+
+               Democrat+GOP+Ideology,data=TAPS,family=binomial)

> summary(ROGLogit)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-0.661  -0.248  -0.206  -0.170   2.962

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -1.9347    1.8114   -1.07   0.285
Education     -0.1185    0.1118   -1.06   0.289
Age10         -0.2107    0.1341   -1.57   0.116
Female         0.2911    0.3966    0.73   0.463
White         0.4348    0.7260    0.60   0.549
Black         1.3503    0.7602    1.78   0.076 .
Asian         1.8616    0.8717    2.14   0.033 *
Democrat      0.2743    0.4999    0.55   0.583
GOP          -0.3170    0.5926   -0.53   0.593
Ideology       0.0217    0.1097    0.20   0.843
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 253.77  on 970  degrees of freedom
Residual deviance: 238.13  on 961  degrees of freedom
AIC: 258.1
```

King/Zeng Rare Events Logit...

```
> devtools::install_github('IQSS/Zelig')
> library(Zelig)

> RElogit<-relogit(RunOutOfGas~Education+Age10+Female+White+Black+Asian+
+                 Democrat+GOP+Ideology, tau = NULL,
+                 case.control = c("prior", "weighting"), # n/a here
+                 bias.correct = TRUE, data=TAPS)

> summary(RElogit)
```

```
Call:
relogit(formula = RunOutOfGas ~ Education + Age10 + Female +
        White + Black + Asian + Democrat + GOP + Ideology, data = TAPS,
        tau = NULL, bias.correct = TRUE, case.control = c("prior",
        "weighting"))
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.7875	1.8114	-0.99	0.324
Education	-0.1173	0.1118	-1.05	0.294
Age10	-0.2076	0.1341	-1.55	0.122
Female	0.2737	0.3966	0.69	0.490
White	0.3916	0.7260	0.54	0.590
Black	1.3600	0.7602	1.79	0.074 .
Asian	1.9579	0.8717	2.25	0.025 *
Democrat	0.2549	0.4999	0.51	0.610
GOP	-0.3057	0.5926	-0.52	0.606
Ideology	0.0272	0.1097	0.25	0.804

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

```
Null deviance: 253.77 on 970 degrees of freedom
Residual deviance: 238.13 on 961 degrees of freedom
AIC: 258.1
```

Number of Fisher Scoring iterations: 7

Firth Logit (for comparison)

```
> relogit.firth<-logistf(RunOutOfGas~Education+Age10+Female+White+Black+Asian+  
+ Democrat+GOP+Ideology,data=TAPS)
```

```
> summary(relogit.firth)
```

```
logistf(formula = RunOutOfGas ~ Education + Age10 + Female +  
White + Black + Asian + Democrat + GOP + Ideology, data = TAPS)
```

Model fitted by Penalized ML

Coefficients:

	coef	se(coef)	lower 0.95	upper 0.95	Chisq	p	method
(Intercept)	-1.7929	1.657	-5.362	1.6045	1.0457	0.3065	2
Education	-0.1167	0.103	-0.331	0.1009	1.1154	0.2909	2
Age10	-0.2071	0.124	-0.469	0.0498	2.4952	0.1142	2
Female	0.2749	0.367	-0.478	1.0490	0.5124	0.4741	2
White	0.3782	0.646	-1.007	1.7513	0.2769	0.5987	2
Black	1.3409	0.677	-0.182	2.7141	2.9875	0.0839	2
Asian	1.9202	0.766	0.149	3.4429	4.4610	0.0347	2
Democrat	0.2550	0.464	-0.688	1.2418	0.2767	0.5989	2
GOP	-0.3061	0.546	-1.479	0.7889	0.2969	0.5858	2
Ideology	0.0267	0.101	-0.191	0.2333	0.0613	0.8044	2

Method: 1-Wald, 2-Profile penalized log-likelihood, 3-None

Likelihood ratio test=17.5 on 9 df, p=0.0415, n=971

Wald test = 318 on 9 df, p = 0

Firth Logit with FLIC

```
> relogit.flic<-logistf(RunOutOfGas~Education+Age10+Female+White+Black+Asian+  
+ Democrat+GOP+Ideology,data=TAPS,flic=TRUE)
```

```
> summary(relogit.flic)
```

```
logistf(formula = RunOutOfGas ~ Education + Age10 + Female +  
White + Black + Asian + Democrat + GOP + Ideology, data = TAPS,  
flic = TRUE)
```

Model fitted by Penalized ML

Coefficients:

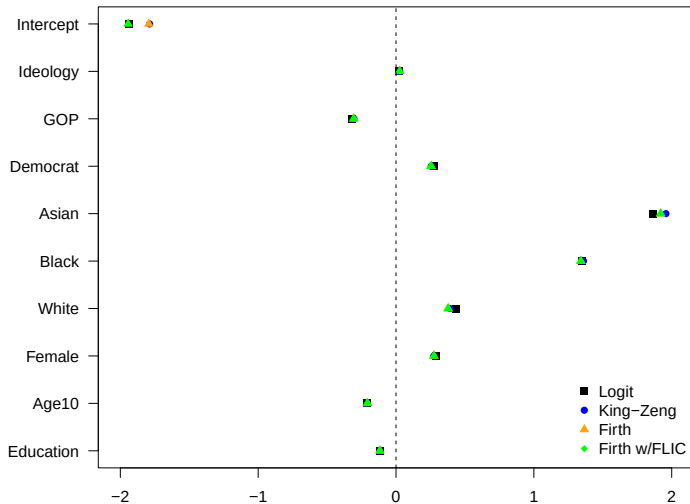
	coef	se(coef)	lower 0.95	upper 0.95	Chisq	p	method
(Intercept)	-1.9430	1.807	-5.486	1.5995	1.0457	0.3065	1
Education	-0.1167	0.112	-0.331	0.1009	1.1154	0.2909	2
Age10	-0.2071	0.134	-0.469	0.0498	2.4952	0.1142	2
Female	0.2749	0.397	-0.478	1.0490	0.5124	0.4741	2
White	0.3782	0.720	-1.007	1.7513	0.2769	0.5987	2
Black	1.3409	0.756	-0.182	2.7141	2.9875	0.0839	2
Asian	1.9202	0.857	0.149	3.4429	4.4610	0.0347	2
Democrat	0.2550	0.501	-0.688	1.2418	0.2767	0.5989	2
GOP	-0.3061	0.590	-1.479	0.7889	0.2969	0.5858	2
Ideology	0.0267	0.110	-0.191	0.2333	0.0613	0.8044	2

Method: 1-Wald, 2-Profile penalized log-likelihood, 3-None

Likelihood ratio test=17.5 on 9 df, p=0.0415, n=971

Wald test = 299 on 9 df, p = 0

Summarizing: $\hat{\beta}$ s



Some Final Thoughts

In no particular order:

- The key to doing King-Zeng is to be able to conduct C-C sampling *in advance*
- BUT: The R implementation of K&Z (in `Zeelig`) is currently a bit buggy (its dependencies are all messed up...)
- In practice: the Firth + FLIC approach is generally superior to King/Zeng (and arguably should *always* be used for binary-response regressions, especially with small-to-medium N s)
- Also: Remember that as your N gets big, the problem goes away; Paul Allison has a (old, but useful) [blog post](#) on that topic.

Other Binary-Response Extensions

Things we'll talk about later:

- Binary responses in panel / longitudinal data
- Multilevel / hierarchical models for binary responses
- Models with (binary) sample selection
- Measurement models for binary outcomes (e.g., item response models)

Things we won't talk about:

- Semi- and non-parametric models (e.g., [Klein and Spady \(1993\)](#), [Christensen \(2024\)](#))
- “Heteroscedastic” models (where $\sigma_i^2 \neq \sigma^2 \forall i$) (see, e.g., Alvarez and Brehm 1995, 1997; Tutz 2018)
- “Bivariate” models, where (e.g.):

$$\{Y_{1i}, Y_{2i}\} \sim BVN(0, 0, 1, 1, \rho)$$

(e.g. [Marra et al. 2023](#))

Nominal Outcomes

Motivation: Discrete *Outcomes*

For J possible outcomes $j \in \{1, 2, \dots, J\}$ on Y :

$$\Pr(Y_i = j) = P_{ij}$$

we require that:

$$\sum_{j=1}^J P_{ij} = 1.$$

To ensure $P_{ij} > 0$:

$$P_{ij} = \exp(\mathbf{X}_i \beta_j).$$

Rescale to make $0 < P_{ij} < 1$:

$$\Pr(Y_i = j) \equiv P_{ij} = \frac{\exp(\mathbf{X}_i \beta_j)}{\sum_{j=1}^J \exp(\mathbf{X}_i \beta_j)}$$

This ensures that:

- $\Pr(Y_i = j) \in (0, 1)$
- $\sum_{j=1}^J \Pr(Y_i = j) = 1.0$

Constrain $\beta_1 = \mathbf{0}$; then:

$$\Pr(Y_i = 1) = \frac{1}{1 + \sum_{j=2}^J \exp(\mathbf{X}_i \beta'_j)}$$

$$\Pr(Y_i = j) = \frac{\exp(\mathbf{X}_i \beta'_j)}{1 + \sum_{j=2}^J \exp(\mathbf{X}_i \beta'_j)}$$

where $\beta'_j = \beta_j - \beta_1$.

This is multinomial logit.

Alternative Motivation: Discrete *Choice*

Utility function for i over j alternatives:

$$U_{ij} = \mu_i + \epsilon_{ij}$$

... varues with $\mathbf{X}$:

$$\mu_i = \mathbf{X}_i \boldsymbol{\beta}_j$$

Yields:

$$\begin{aligned} \Pr(Y_i = j) &= \Pr(U_{ij} > U_{i\ell} \forall \ell \neq j \in J) \\ &= \Pr(\mu_i + \epsilon_{ij} > \mu_i + \epsilon_{i\ell} \forall \ell \neq j \in J) \\ &= \Pr(\mathbf{X}_i \boldsymbol{\beta}_j + \epsilon_{ij} > \mathbf{X}_i \boldsymbol{\beta}_\ell + \epsilon_{i\ell} \forall \ell \neq j \in J) \\ &= \Pr(\epsilon_{ij} - \epsilon_{i\ell} > \mathbf{X}_i \boldsymbol{\beta}_\ell - \mathbf{X}_i \boldsymbol{\beta}_j \forall \ell \neq j \in J) \end{aligned}$$

$\epsilon \sim ???$

- *Type I Extreme Value*
- Density: $f(\epsilon) = \exp[-\epsilon - \exp(-\epsilon)]$
- CDF: $\int f(\epsilon) \equiv F(\epsilon) = \exp[-\exp(-\epsilon)]$
- $\rightarrow$ Multinomial Logit

The probability of choosing choice j is:

$$\begin{aligned}
 \Pr(Y_i = j) &= \Pr(U_j > U_1, U_j > U_2, \dots, U_j > U_J) \\
 &= \int f(\epsilon_j) \left[\int_{-\infty}^{\epsilon_{ij} + \mathbf{X}_i \beta_j - \mathbf{X}_i \beta_1} f(\epsilon_1) d\epsilon_1 \times \int_{-\infty}^{\epsilon_{ij} + \mathbf{X}_i \beta_j - \mathbf{X}_i \beta_2} f(\epsilon_2) d\epsilon_2 \times \dots \right] d\epsilon_j \\
 &= \int f(\epsilon_j) \times \exp[-\exp(\epsilon_{ij} + \mathbf{X}_i \beta_j - \mathbf{X}_i \beta_1)] \times \\
 &\quad \exp[-\exp(\epsilon_{ij} + \mathbf{X}_i \beta_j - \mathbf{X}_i \beta_2)] \times \dots d\epsilon_j \\
 &= \frac{\exp(\mathbf{X}_i \beta_j)}{\sum_{j=1}^J \exp(\mathbf{X}_i \beta_j)}
 \end{aligned}$$

Define:

$$\begin{aligned}\delta_{ij} &= 1 \text{ if } Y_i = j, \\ &= 0 \text{ otherwise.}\end{aligned}$$

Then:

$$\begin{aligned}L_i &= \prod_{j=1}^J [\Pr(Y_i = j)]^{\delta_{ij}} \\ &= \prod_{j=1}^J \left[\frac{\exp(\mathbf{X}_i \beta_j)}{\sum_{j=1}^J \exp(\mathbf{X}_i \beta_j)} \right]^{\delta_{ij}}\end{aligned}$$

So:

$$L = \prod_{i=1}^N \prod_{j=1}^J \left[\frac{\exp(\mathbf{X}_i \beta_j)}{\sum_{j=1}^J \exp(\mathbf{X}_i \beta_j)} \right]^{\delta_{ij}}$$

and:

$$\ln L = \sum_{i=1}^N \sum_{j=1}^J \delta_{ij} \ln \left[\frac{\exp(\mathbf{X}_i \beta_j)}{\sum_{j=1}^J \exp(\mathbf{X}_i \beta_j)} \right]$$

...which can be estimated in the usual way(s).

Example: The 1992 U.S. Presidential Election



1992 American National Election Study

Data:

- Y (PresVote) $\in \{\text{Bush}(= 1), \text{Clinton}(= 2), \text{Perot}(= 3)\}$
- $\mathbf{X}$ = political demographic characteristics + “feeling thermometers”

```
> describe(NES92)
```

	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
ID	1	1473	4671.15	1104.02	5113	4681.28	1546.35	3001	6251	3250	-0.11	-1.66	28.77
VotedFor*	2	1473	1.85	0.71	2	1.82	1.48	1	3	2	0.22	-1.03	0.02
PresVote	3	1473	1.85	0.71	2	1.82	1.48	1	3	2	0.22	-1.03	0.02
PartyID	4	1473	3.75	2.11	3	3.69	2.97	1	7	6	0.15	-1.39	0.06
Age	5	1473	45.89	16.67	43	44.85	17.79	18	91	73	0.50	-0.72	0.43
FamIncome	6	1473	15.53	5.76	16	16.10	5.93	1	24	23	-0.78	-0.17	0.15
Female	7	1473	0.51	0.50	1	0.52	0.00	0	1	1	-0.06	-2.00	0.01
White*	8	1473	1.88	0.33	2	1.97	0.00	1	2	1	-2.31	3.36	0.01
FT.Bush	9	1473	51.75	27.26	60	52.99	29.65	0	100	100	-0.30	-0.72	0.71
FT.Clinton	10	1473	55.77	25.08	60	57.35	29.65	0	100	100	-0.45	-0.37	0.65
FT.Perot	11	1473	44.85	26.51	50	44.90	29.65	0	100	100	-0.16	-0.68	0.69

Model:

$$\text{PresVote}_i = f(\beta_0 + \beta_1 \times \text{PartyID}_i + \beta_2 \times \text{Age}_i + \beta_3 \times \text{White}_i + \beta_4 \times \text{Female}_i)$$

MNL #1, using vglm (“Baseline” = Perot)

```
> NES92.mlogit<-vglm(PresVote~PartyID+Age+White+Female,multinomial,data=NES92)
> summary(NES92.mlogit)
```

Call:

```
vglm(formula = PresVote ~ PartyID + Age + White + Female, family = multinomial,
      data = NES92)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept):1	-1.98008	0.52454	-3.77	0.00016	***
(Intercept):2	3.82657	0.46402	8.25	< 2e-16	***
PartyID:1	0.50132	0.04870	10.29	< 2e-16	***
PartyID:2	-0.63429	0.04918	-12.90	< 2e-16	***
Age:1	0.01556	0.00504	3.09	0.00203	**
Age:2	0.01296	0.00510	2.54	0.01096	*
WhiteWhite:1	-0.87918	0.43605	-2.02	0.04377	*
WhiteWhite:2	-1.86826	0.38611	-4.84	0.0000013	***
Female:1	0.50928	0.16266	3.13	0.00174	**
Female:2	0.38427	0.16267	2.36	0.01816	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,3]), log(mu[,2]/mu[,3])

Residual deviance: 2107 on 2936 degrees of freedom

Log-likelihood: -1054 on 2936 degrees of freedom

Number of Fisher scoring iterations: 5

No Hauck-Donner effect found in any of the estimates

Reference group is level 3 of the response

MNL #2, using multinom ("Baseline" = Perot)

```
> NES92$PresVote2<-factor(NES92$PresVote,
+                           levels = c("3", "1", "2"),
+                           labels = c("Perot", "Bush", "Clinton"))

> NES92.mlogit2<-multinom(PresVote2~PartyID+Age+White+Female,data=NES92)
# weights:  18 (10 variable)
initial value 1618.255901
iter  10 value 1080.908630
final value 1053.650588
converged

> summary(NES92.mlogit2)
Call:
multinom(formula = PresVote2 ~ PartyID + Age + White + Female,
          data = NES92)

Coefficients:
      (Intercept) PartyID      Age WhiteWhite Female
Bush           -1.98   0.501 0.0156    -0.879   0.509
Clinton         3.83  -0.634 0.0130    -1.868   0.384

Std. Errors:
      (Intercept) PartyID      Age WhiteWhite Female
Bush           0.525  0.0487 0.00504      0.436   0.163
Clinton        0.464  0.0492 0.00510      0.386   0.163

Residual Deviance: 2107
AIC: 2127
```

MNL #3, using mlogit

First, we have to “reshape” the data:

```
> head(NES92)
  ID VotedFor PresVote PartyID Age FamIncome Female      White FT.Bush FT.Clinton FT.Perot PresVote2
1 3001     Bush        1      6  31          20      0    White      85         30         0      Bush
2 3002     Bush        1      7  89           9      1    White     100          0         0      Bush
3 3003     Bush        1      7  35          17      1    White      85         30        60      Bush
4 3005 Clinton        2      6  27           3      1 Non-White     40         60        60  Clinton
5 3006 Clinton        2      2  54          15      1    White      30         70        50  Clinton
6 3007 Clinton        2      1  45           2      1 Non-White     15         70        50  Clinton
```

```
> AltNES92<-dfidx(NES92,varying=9:11,shape="wide",choice="VotedFor")
```

```
> head(AltNES92)
~~~~~
first 10 observations out of 4419
~~~~~
  ID VotedFor PresVote PartyID Age FamIncome Female      White PresVote2 FT      idx
1 3001     TRUE        1      6  31          20      0    White      Bush  85 1:Bush
2 3001    FALSE        1      6  31          20      0    White      Bush  30 1:nton
3 3001    FALSE        1      6  31          20      0    White      Bush   0 1:erot
4 3002     TRUE        1      7  89           9      1    White      Bush 100 2:Bush
5 3002    FALSE        1      7  89           9      1    White      Bush   0 2:nton
6 3002    FALSE        1      7  89           9      1    White      Bush   0 2:erot
7 3003     TRUE        1      7  35          17      1    White      Bush  85 3:Bush
8 3003    FALSE        1      7  35          17      1    White      Bush  30 3:nton
9 3003    FALSE        1      7  35          17      1    White      Bush  60 3:erot
```

MNL #3, using mlogit (continued)

Now, fit the model:

```
> NES92.mlogit3<-mlogit(VotedFor~0|PartyID+Age+White+Female,data=AltNES92,reflevel="Perot")
> summary(NES92.mlogit3)
```

```
Call:
mlogit(formula = VotedFor ~ 0 | PartyID + Age + White + Female,
      data = AltNES92, reflevel = "Perot", method = "nr")
```

Frequencies of alternatives:choice

	Perot	Bush	Clinton
	0.191	0.339	0.469

```
nr method
5 iterations, 0h:0m:0s
g'(-H)^-1g = 4.94E-08
gradient close to zero
```

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept):Bush	-1.98008	0.52454	-3.77	0.00016 ***
(Intercept):Clinton	3.82657	0.46403	8.25	2.2e-16 ***
PartyID:Bush	0.50132	0.04870	10.29	< 2e-16 ***
PartyID:Clinton	-0.63429	0.04918	-12.90	< 2e-16 ***
Age:Bush	0.01556	0.00504	3.09	0.00203 **
Age:Clinton	0.01296	0.00510	2.54	0.01096 *
WhiteWhite:Bush	-0.87918	0.43606	-2.02	0.04378 *
WhiteWhite:Clinton	-1.86826	0.38612	-4.84	1.3e-06 ***
Female:Bush	0.50928	0.16266	3.13	0.00174 **
Female:Clinton	0.38427	0.16267	2.36	0.01816 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -1050

McFadden R²: 0.311

Likelihood ratio test : chisq = 952 (p.value = <2e-16)

MNL: 1992 Election (“Baseline” = Bush)

```
> Bush.nes92.mlogit<-vglm(PresVote~PartyID+Age+White+Female,  
+                           data=NES92,family=multinomial(refLevel=1))  
> summary(Bush.nes92.mlogit)
```

Call:

```
vglm(formula = PresVote ~ PartyID + Age + White + Female, family = multinomial(refLevel = 1),  
     data = NES92)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept):1	5.80665	0.44301	13.11	< 2e-16 ***
(Intercept):2	1.98008	0.52454	3.77	0.00016 ***
PartyID:1	-1.13561	0.05486	-20.70	< 2e-16 ***
PartyID:2	-0.50132	0.04870	-10.29	< 2e-16 ***
Age:1	-0.00260	0.00514	-0.51	0.61276
Age:2	-0.01556	0.00504	-3.09	0.00203 **
WhiteWhite:1	-0.98908	0.31346	-3.16	0.00160 **
WhiteWhite:2	0.87918	0.43605	2.02	0.04377 *
Female:1	-0.12500	0.16895	-0.74	0.45936
Female:2	-0.50928	0.16266	-3.13	0.00174 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,2]/mu[,1]), log(mu[,3]/mu[,1])

Residual deviance: 2107 on 2936 degrees of freedom

Log-likelihood: -1054 on 2936 degrees of freedom

Number of Fisher scoring iterations: 5

No Hauck-Donner effect found in any of the estimates

Reference group is level 1 of the response

MNL: 1992 Election (“Baseline” = Clinton)

```
> Clinton.nes92.mlogit<-vglm(PresVote~PartyID+Age+White+Female,  
+                             data=NES92,family=multinomial(refLevel=2))  
> summary(Clinton.nes92.mlogit)
```

Call:

```
vglm(formula = PresVote ~ PartyID + Age + White + Female, family = multinomial(refLevel = 2),  
     data = NES92)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept):1	-5.80665	0.44301	-13.11	< 2e-16 ***
(Intercept):2	-3.82657	0.46402	-8.25	< 2e-16 ***
PartyID:1	1.13561	0.05486	20.70	< 2e-16 ***
PartyID:2	0.63429	0.04918	12.90	< 2e-16 ***
Age:1	0.00260	0.00514	0.51	0.6128
Age:2	-0.01296	0.00510	-2.54	0.0110 *
WhiteWhite:1	0.98908	0.31346	3.16	0.0016 **
WhiteWhite:2	1.86826	0.38611	4.84	0.0000013 ***
Female:1	0.12500	0.16895	0.74	0.4594
Female:2	-0.38427	0.16267	-2.36	0.0182 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,1]/mu[,2]), log(mu[,3]/mu[,2])

Residual deviance: 2107 on 2936 degrees of freedom

Log-likelihood: -1054 on 2936 degrees of freedom

Number of Fisher scoring iterations: 5

Warning: Hauck-Donner effect detected in the following estimate(s):

'(Intercept):2'

Reference group is level 2 of the response

PartyID Coefficient Estimates and “Baselines”

Note: PartyID is 1 (strong Democrat) $\rightarrow$ 7 (strong Republican)

		<u>“Baseline” category</u>		
		Clinton	Perot	Bush
Comparison	Clinton	–	-0.63	-1.14
Category	Perot	0.63	–	-0.50
	Bush	1.14	0.50	–

Consider the choice of Bush vs. Perot:

```
> NES92$PickBush<-NA
> NES92$PickBush<-ifelse(NES92$VotedFor=="Bush",1,NES92$PickBush)
> NES92$PickBush<-ifelse(NES92$VotedFor=="Perot",0,NES92$PickBush)
> BushBinary<-glm(PickBush~PartyID+Age+White+Female,data=NES92,family="binomial")
> summary(BushBinary)
```

Call:

```
glm(formula = PickBush ~ PartyID + Age + White + Female, family = "binomial",
    data = NES92)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.9024	0.5372	-3.54	0.00040	***
PartyID	0.5106	0.0505	10.12	< 2e-16	***
Age	0.0143	0.0052	2.75	0.00595	**
WhiteWhite	-0.9817	0.4586	-2.14	0.03230	*
Female	0.5768	0.1683	3.43	0.00061	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1022.50 on 781 degrees of freedom
Residual deviance: 880.28 on 777 degrees of freedom
(691 observations deleted due to missingness)
AIC: 890.3

Number of Fisher Scoring iterations: 4

MNL and Binary Logit (continued)

What about Clinton vs. Perot?:

```
> NES92$PickClinton<-NA
> NES92$PickClinton<-ifelse(NES92$VotedFor=="Clinton",1,NES92$PickClinton)
> NES92$PickClinton<-ifelse(NES92$VotedFor=="Perot",0,NES92$PickClinton)
> ClintonBinary<-glm(PickClinton~PartyID+Age+White+Female,data=NES92,family="binomial")
> summary(ClintonBinary)
```

Call:

```
glm(formula = PickClinton ~ PartyID + Age + White + Female, family = "binomial",
    data = NES92)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.92614	0.48490	8.10	5.6e-16 ***
PartyID	-0.68125	0.05301	-12.85	< 2e-16 ***
Age	0.01381	0.00537	2.57	0.0101 *
WhiteWhite	-1.91056	0.39879	-4.79	1.7e-06 ***
Female	0.51690	0.17024	3.04	0.0024 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

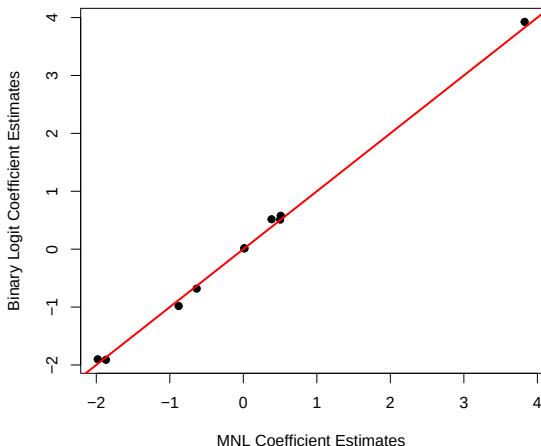
(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1171.48 on 972 degrees of freedom
Residual deviance: 861.57 on 968 degrees of freedom
(500 observations deleted due to missingness)
AIC: 871.6

Number of Fisher Scoring iterations: 5

MNL and Binary Logit (continued)

Are the $\hat{\beta}$ s the same? (A: Yes, basically...)



It is exactly the same as the multinomial logit model. Period.

Choice-Specific Covariates: Data Structure

```
> head(AltNES92)
```

	ID	VotedFor	PresVote	PartyID	Age	FamIncome	Female	White	PresVote2	FT	idx\$id1	\$id2
*	<dbl>	<lgl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<fct>	<dbl>	<int>	<fct>
1	3001	TRUE	1	6	31	20	0	White	Bush	85	1	Bush
2	3001	FALSE	1	6	31	20	0	White	Bush	30	1	Clinton
3	3001	FALSE	1	6	31	20	0	White	Bush	0	1	Perot
4	3002	TRUE	1	7	89	9	1	White	Bush	100	2	Bush
5	3002	FALSE	1	7	89	9	1	White	Bush	0	2	Clinton
6	3002	FALSE	1	7	89	9	1	White	Bush	0	2	Perot
7	3003	TRUE	1	7	35	17	1	White	Bush	85	3	Bush
8	3003	FALSE	1	7	35	17	1	White	Bush	30	3	Clinton
9	3003	FALSE	1	7	35	17	1	White	Bush	60	3	Perot
10	3005	FALSE	2	6	27	3	1	Non-White	Clinton	40	4	Bush

4,409 more rows
Use 'print(n = ...)' to see more rows

Note that:

$$\Pr(Y_{ij} = 1) = \frac{\exp(\mathbf{Z}_{ij}\gamma)}{\sum_{j=1}^J \exp(\mathbf{Z}_{ij}\gamma)}$$

Combinations: $\mathbf{X}_i\beta$ and $\mathbf{Z}_{ij}\gamma$:

- “Fixed effects” (choice-specific intercepts), plus
- Observation-specific $\mathbf{X}$ s, plus
- Interactions...

CL in R (Feeling Thermometers only)

```
> NES92.clogit<-mlogit(VotedFor~FT,data=AltNES92,reflevel="Perot")
> summary(NES92.clogit)
```

Call:

```
mlogit(formula = VotedFor ~ FT, data = AltNES92, reflevel = "Perot",
       method = "nr")
```

Frequencies of alternatives:choice

	Perot	Bush	Clinton
	0.191	0.339	0.469

nr method

6 iterations, 0h:0m:0s

g'(-H)⁻¹g = 0.00219

successive function values within tolerance limits

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept):Bush	0.03307	0.10039	0.33	0.74
(Intercept):Clinton	0.45841	0.09253	4.95	0.00000073 ***
FT	0.07512	0.00314	23.89	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -801

McFadden R²: 0.476

Likelihood ratio test : chisq = 1460 (p.value = <2e-16)

CL in R ("Full" Model)

```
> NES92.clogit2<-mlogit(VotedFor~FT|PartyID+Age+White+Female,data=AltNES92,reflevel="Perot")
> summary(NES92.clogit2)
```

Frequencies of alternatives:choice

Perot	Bush	Clinton
0.191	0.339	0.469

nr method

6 iterations, 0h:0m:0s

g'(-H)^-1g = 3.06E-08

gradient close to zero

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept):Bush	-0.39416	0.58730	-0.67	0.50214
(Intercept):Clinton	2.69235	0.50403	5.34	9.2e-08 ***
FT	0.06231	0.00325	19.17	< 2e-16 ***
PartyID:Bush	0.22298	0.05842	3.82	0.00014 ***
PartyID:Clinton	-0.40620	0.05798	-7.01	2.4e-12 ***
Age:Bush	0.00868	0.00612	1.42	0.15639
Age:Clinton	0.00839	0.00598	1.40	0.16040
WhiteWhite:Bush	-1.31961	0.47725	-2.77	0.00569 **
WhiteWhite:Clinton	-1.57156	0.41404	-3.80	0.00015 ***
Female:Bush	0.39271	0.19995	1.96	0.04953 *
Female:Clinton	0.28585	0.19474	1.47	0.14213

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Log-Likelihood: -723

McFadden R^2: 0.527

Likelihood ratio test : chisq = 1610 (p.value = <2e-16)

Interpretation: Baseline MNL Results

```
> NES.MNL<-vglm(PresVote~PartyID+Age+White+Female,data=NES92,  
+ multinomial(refLevel=1)) # Bush is comparison category  
> summaryvglm(NES.MNL)
```

Call:

```
vglm(formula = PresVote ~ PartyID + Age + White + Female, family = multinomial(refLevel = 1),  
data = NES92)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept):1	5.80665	0.44301	13.11	< 2e-16	***
(Intercept):2	1.98008	0.52454	3.77	0.00016	***
PartyID:1	-1.13561	0.05486	-20.70	< 2e-16	***
PartyID:2	-0.50132	0.04870	-10.29	< 2e-16	***
Age:1	-0.00260	0.00514	-0.51	0.61276	
Age:2	-0.01556	0.00504	-3.09	0.00203	**
WhiteWhite:1	-0.98908	0.31346	-3.16	0.00160	**
WhiteWhite:2	0.87918	0.43605	2.02	0.04377	*
Female:1	-0.12500	0.16895	-0.74	0.45936	
Female:2	-0.50928	0.16266	-3.13	0.00174	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Names of linear predictors: log(mu[,2]/mu[,1]), log(mu[,3]/mu[,1])

Residual deviance: 2107 on 2936 degrees of freedom

Log-likelihood: -1054 on 2936 degrees of freedom

Number of Fisher scoring iterations: 5

No Hauck-Donner effect found in any of the estimates

Reference group is level 1 of the response

Pretty Things Using modelsummary

```
> mod<-list(NES92.mlogit2)
```

```
> get_estimates(mod[[1]])
```

	term	estimate	std.error	conf.level	conf.low	conf.high	statistic	df.error	p.value	response	s.value	group
1	(Intercept)	-1.9797	0.52455	0.95	-3.00781	-0.9516	-3.77	Inf	1.61e-04	Bush	12.6	
2	PartyID	0.5013	0.04870	0.95	0.40586	0.5967	10.29	Inf	7.47e-25	Bush	80.1	
3	Age	0.0156	0.00504	0.95	0.00568	0.0254	3.09	Inf	2.03e-03	Bush	8.9	
4	White	-0.8794	0.43607	0.95	-1.73404	-0.0247	-2.02	Inf	4.37e-02	Bush	4.5	
5	Female	0.5093	0.16266	0.95	0.19047	0.8281	3.13	Inf	1.74e-03	Bush	9.2	
6	(Intercept)	3.8267	0.46404	0.95	2.91718	4.7362	8.25	Inf	1.63e-16	Clinton	52.4	
7	PartyID	-0.6343	0.04918	0.95	-0.73067	-0.5379	-12.90	Inf	4.71e-38	Clinton	124.0	
8	Age	0.0130	0.00510	0.95	0.00297	0.0229	2.54	Inf	1.10e-02	Clinton	6.5	
9	White	-1.8683	0.38613	0.95	-2.62512	-1.1115	-4.84	Inf	1.31e-06	Clinton	19.5	
10	Female	0.3843	0.16267	0.95	0.06544	0.7031	2.36	Inf	1.82e-02	Clinton	5.8	

```
> modelsummary(mod, shape=term+response~statistic)
```

(1)		
	response	Est. S.E.
(Intercept)	Bush	-1.980 0.525
	Clinton	3.827 0.464
PartyID	Bush	0.501 0.049
	Clinton	-0.634 0.049
Age	Bush	0.016 0.005
	Clinton	0.013 0.005
White/White	Bush	-0.879 0.436
	Clinton	-1.868 0.386
Female	Bush	0.509 0.163
	Clinton	0.384 0.163
Num Obs		1473
R2		0.311
R2 Adj		0.310
AIC		7177.3
BIC		7180.3
RMSE		0.37

```
> modelsummary(mod, shape=model+term~response)
```

		Bush	Clinton
(1)	(Intercept)	-1.980	3.827
		(0.525)	(0.464)
PartyID		0.501	-0.634
		(0.049)	(0.049)
Age		0.016	0.013
		(0.005)	(0.005)
White/White		-0.879	-1.868
		(0.436)	(0.386)
Female		0.509	0.384
		(0.163)	(0.163)

Global In LR statistic Q tests:

$$\hat{\beta} = \mathbf{0} \forall j, k$$

$$Q \sim \chi^2_{(J-1)(k-1)}$$

Test H: No Effect of Age

Is the effect of Age across the three candidates equal to zero?

```
> library(aod)
> wald.test(b=c(t(coef(NES.MNL))),Sigma=vcov(NES.MNL),Terms=c(5,6))
```

Wald test:

Chi-squared test:

X2 = 11.0, df = 2, P(> X2) = 0.0042

Test H: No Difference – Clinton vs. Bush

Are the estimated coefficients for Clinton (vs. Bush) jointly equal to zero?

```
> wald.test(b=c(t(coef(NES.MNL))),Sigma=vcov(NES.MNL),Terms=c(1,3,5,7,9))
```

Wald test:

Chi-squared test:

$X^2 = 444.6$, $df = 5$, $P(> X^2) = 0.0$

Interpretation: Marginal Effects

Marginal effects are:

$$\frac{\partial \Pr(Y_i = j)}{\partial X_k} = \Pr(Y_i = j | \mathbf{X}) \left[\hat{\beta}_{jk} - \sum_{j=1}^J \hat{\beta}_{jk} \times \Pr(Y_i = j | \mathbf{X}) \right]$$

Depend on:

- $\widehat{\Pr(Y_i = j)}$
- $\hat{\beta}_{jk}$
- $\sum_{j=1}^J \hat{\beta}_{jk}$

Available for `-multinom-` (in the `-nnet-` package) via the `-margins-` package...

Marginal Effects: Illustrated

```
> MNL.alt<-multinom(PresVote2~PartyID+Age+White+Female,data=NES92,Hess=TRUE)
# weights:  18 (10 variable)
initial  value 1618.255901
iter    10 value 1080.908630
final    value 1053.650588
converged
```

```
> summary(marginal_effects(MNL.alt))
```

dydx_PartyID	dydx_Age	dydx_Female	dydx_WhiteWhite
Min. :-0.0908	Min. :-0.00362	Min. :-0.1158	Min. :0.0416
1st Qu.: -0.0439	1st Qu.: -0.00282	1st Qu.: -0.0892	1st Qu.: 0.0943
Median : 0.0185	Median :-0.00215	Median :-0.0674	Median :0.1352
Mean : 0.0083	Mean :-0.00207	Mean :-0.0648	Mean :0.1435
3rd Qu.: 0.0618	3rd Qu.: -0.00138	3rd Qu.: -0.0420	3rd Qu.:0.1848
Max. : 0.1070	Max. :-0.00011	Max. :-0.0034	Max. :0.2926

Odds (“Relative Risk”) Ratios

MNL has:

$$\ln \left[\frac{\Pr(Y_i = j | \mathbf{X})}{\Pr(Y_i = j' | \mathbf{X})} \right] = \mathbf{X}(\hat{\beta}_j - \hat{\beta}_{j'})$$

Setting $\hat{\beta}_{j'} = \mathbf{0}$:

$$\ln \left[\frac{\Pr(Y_i = j | \mathbf{X})}{\Pr(Y_i = j' | \mathbf{X})} \right] = \mathbf{X}\hat{\beta}_j$$

One-Unit Change in X_k :

$$RRR_{jk} = \exp(\beta_{jk})$$

δ -Unit Change in X_k :

$$RRR_{jk} = \exp(\beta_{jk} \times \delta)$$

Odds (“Relative Risk”) Ratios

```
> mnl.or <- function(model) {  
  coeffs <- c(t(coef(NES.MNL)))  
  lci <- exp(coeffs - 1.96 * diag(vcov(NES.MNL))^0.5)  
  or <- exp(coeffs)  
  uci <- exp(coeffs + 1.96* diag(vcov(NES.MNL))^0.5)  
  lreg.or <- cbind(lci, or, uci)  
  lreg.or  
}
```

```
> mnl.or(NES.MNL)
```

	lci	or	uci
(Intercept):1	139.540	332.504	792.309
(Intercept):2	2.591	7.243	20.250
PartyID:1	0.288	0.321	0.358
PartyID:2	0.551	0.606	0.666
Age:1	0.987	0.997	1.008
Age:2	0.975	0.985	0.994
WhiteWhite:1	0.201	0.372	0.688
WhiteWhite:2	1.025	2.409	5.662
Female:1	0.634	0.882	1.229
Female:2	0.437	0.601	0.827

Odds Ratios: Interpretation

Odds ratio interpretations:

- A one unit increase in **partyid** corresponds to:
 - A decrease in the odds of a Clinton vote, versus a vote for Bush, of $\exp(-1.136) = 0.321$ (or about 68 percent), and
 - A decrease in the odds of a Perot vote, versus a vote for Bush, of $\exp(-0.501) = 0.606$ (or about 40 percent).
 - These are *large* decreases in the odds – not surprisingly, more Republican voters are *much* more likely to vote for Bush than for Perot or Clinton.
- Similarly, **female** voters are:
 - No more or less likely to vote for Clinton vs. Bush (OR=0.88), but
 - Roughly 40 percent less likely to have voted for Perot (OR=0.60).

$$\begin{aligned}\Pr(\widehat{\text{presvote}}_i = \text{Bush}) &= \frac{\exp(\mathbf{X}_i \hat{\beta}_{\text{Bush}})}{\sum_{j=1}^J \exp(\mathbf{X}_i \hat{\beta}_j)} \\ &= \frac{1}{1 + \sum_{j=2}^J \exp(\mathbf{X}_i \hat{\beta}_j)}\end{aligned}$$

In-Sample Predicted Outcomes

Generate predicted vote choices:

```
> NES92$Predictions<-" "  
> NES92$Predictions<-ifelse(fitted.values(NES.MNL)[,1]>fitted.values(NES.MNL)[,2]  
+ & fitted.values(NES.MNL)[,1]>fitted.values(NES.MNL)[,3],  
+ paste("Bush"),NES92$Predictions) # Bush  
> NES92$Predictions<-ifelse(fitted.values(NES.MNL)[,2]>fitted.values(NES.MNL)[,1]  
+ & fitted.values(NES.MNL)[,2]>fitted.values(NES.MNL)[,3],  
+ paste("Clinton"),NES92$Predictions) # Clinton  
> NES92$Predictions<-ifelse(fitted.values(NES.MNL)[,3]>fitted.values(NES.MNL)[,1]  
+ & fitted.values(NES.MNL)[,3]>fitted.values(NES.MNL)[,2],  
+ paste("Perot"),NES92$Predictions) # Perot  
  
> # "Confusion Table":  
>  
> table(NES92$VotedFor,NES92$Predictions)
```

	Bush	Clinton	Perot
Bush	415	77	8
Clinton	56	619	16
Perot	135	133	14

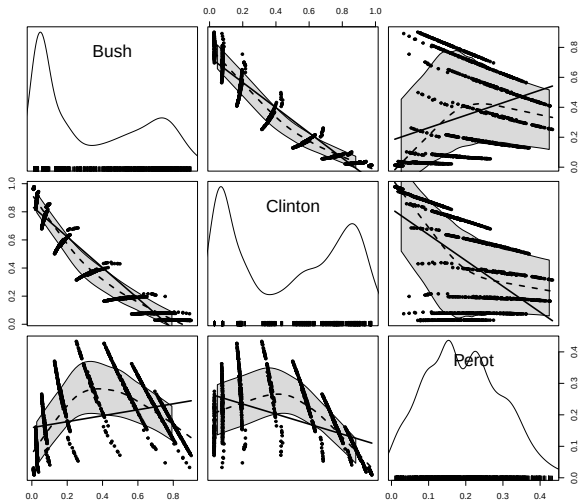
Model fit:

- “Null” Model: $\left(\frac{691}{1473}\right) = 46.9\%$ correct.
- Estimated model: $\frac{(415+619+14)}{1473} = \frac{1048}{1473} = 71.2\%$ correct.
- $PRE = \frac{1048-691}{1473-691} = \frac{357}{782} = 45.7\%$.
- Correct predictions: 90% Clinton, 83% Bush, 5% Perot.

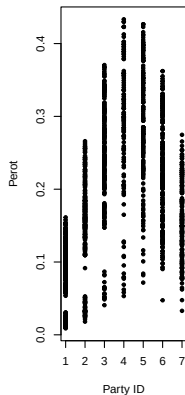
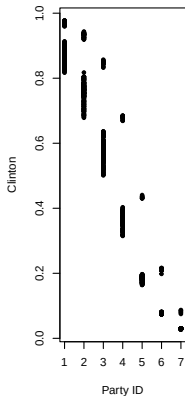
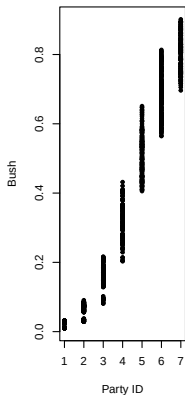
In-Sample Predicted Probabilities

```
> hats<-as.data.frame(fitted.values(NES.MNL))
> names(hats)[3]<-"Perot" # nice names...
> names(hats)[2]<-"Clinton"
> names(hats)[1]<-"Bush"
> attach(hats)

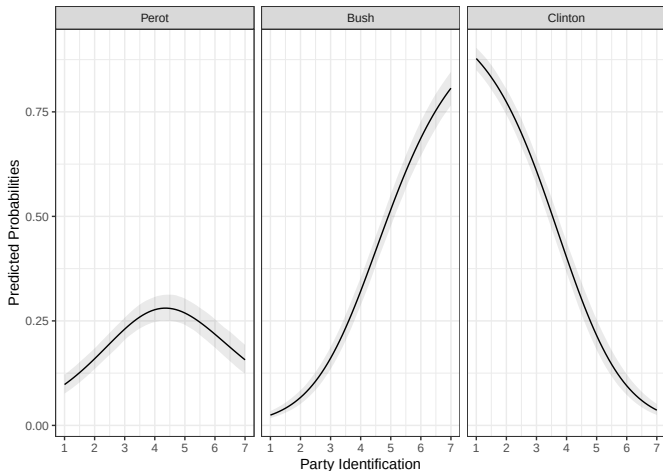
> library(car)
> scatterplot.matrix(~Bush+Clinton+Perot,
  diagonal="histogram",col=c("black","grey"))
```



In-Sample $\hat{P}$ rs vs. partyid

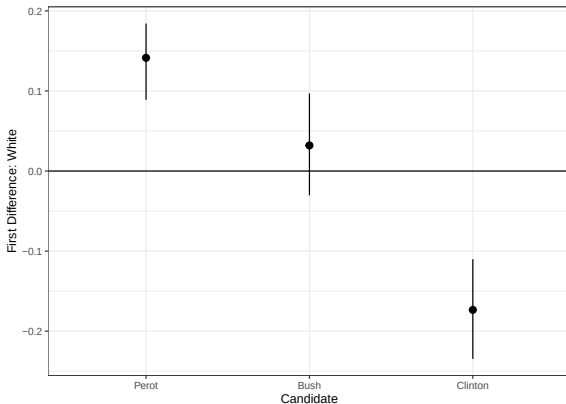


Out-Of-Sample Predictions (using MNLpred)



OOS First Differences (using MNLpred)

First differences in probabilities associated with White:



Conditional Logit: Example

```
> summary(NES92.clogit2)
```

Call:

```
mlogit(formula = VotedFor ~ FT | PartyID + Age + White + Female,  
data = AltNES92, reflevel = "Perot", method = "nr")
```

Frequencies of alternatives:choice

```
Perot    Bush Clinton  
0.191    0.339    0.469
```

nr method

6 iterations, 0h:0m:0s

g'(-H)^-1g = 3.06E-08

gradient close to zero

Coefficients :

	Estimate	Std. Error	z-value	Pr(> z)
(Intercept):Bush	-0.39416	0.58730	-0.67	0.50214
(Intercept):Clinton	2.69235	0.50403	5.34	9.2e-08 ***
FT	0.06231	0.00325	19.17	< 2e-16 ***
PartyID:Bush	0.22298	0.05842	3.82	0.00014 ***
PartyID:Clinton	-0.40620	0.05798	-7.01	2.4e-12 ***
Age:Bush	0.00868	0.00612	1.42	0.15639
Age:Clinton	0.00839	0.00598	1.40	0.16040
WhiteWhite:Bush	-1.31961	0.47725	-2.77	0.00569 **
WhiteWhite:Clinton	-1.57156	0.41404	-3.80	0.00015 ***
Female:Bush	0.39271	0.19995	1.96	0.04953 *
Female:Clinton	0.28585	0.19474	1.47	0.14213

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

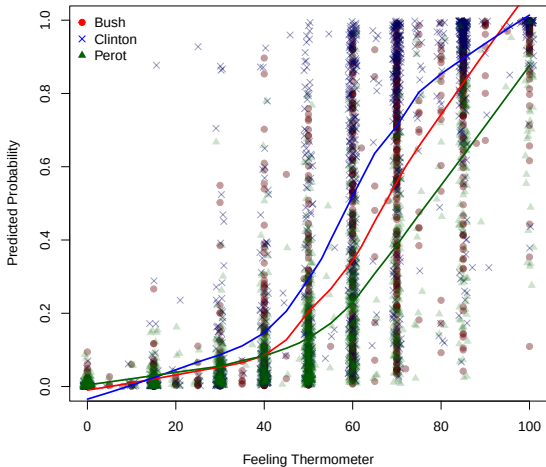
Log-Likelihood: -723

McFadden R²: 0.527

Likelihood ratio test : chisq = 1610 (p.value = <2e-16)

Conditional Logit: In-Sample Predicted Probabilities

```
> CLhats<-predict(NES92.clogit2,AltNES92)
```



Various things:

- “Independence of Irrelevant Alternatives”
- → Multinomial Probit
- → Heteroscedastic Extreme Value model
- “Mixed” Logit
- Nested Logit, etc.